

Challenge 3: Caccetta–Häggkvist Conjecture

Given a simple digraph, if each vertex has an out-neighbour, then it contains a directed cycle [1], but not necessarily a short one. As the minimum out-degree increases, girth is expected to decrease. A natural construction is the digraph on n cyclically ordered vertices, where each vertex is joined to the next r vertices. In this example, the minimum out-degree is r and the girth is $\lceil n/r \rceil$. The Caccetta–Häggkvist conjecture asserts that no digraph with minimum out-degree r can do better.

Conjecture 1.1 (Caccetta–Häggkvist Conjecture [2]). Let $r \in \mathbb{N}$ with $r \geq 1$. Every simple directed graph on n vertices with minimum out-degree at least r has a cycle of length at most $\lceil n/r \rceil$.

For $r = 1$ the statement is elementary. The conjecture was proved for $r = 2$ by Caccetta and Häggkvist [2], for $r = 3$ by Hamidoune [3], and for $r = 4$ and $r = 5$ by Hoàng and Reed [4]. Shen proved it for every r under the additional assumption $n \geq 2r^2 - 3r + 1$ [5].

References

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